

SRW3 Phase 1G — Authority-Rooted State and Execution Evidence

****Status:** PHASE 1G EXECUTED — the authority boundary is answered with a qualified, boundary-honest result: authority is INHERITED, NOT CREATED. The non-circular authority machinery is PROVED (branch-wise + accept direction), the authority graph is mechanically acyclic, every self-authorizing / substitution attack in the CM-G set is REJECTED at an exact diagnosed layer, and the smallest authoritative integration point is DEMONSTRATED over real EVM execution (Mode A). The protocol root's own authority is the explicit, structured assumption A-G2 that replaces A-E3 — in deployment it is the Engine-API boundary and remains REQUIRES CLIENT/PROTOCOL SUPPORT.**

This report was produced from the phase1g branch derived from the Phase 1F closure commit (b6c651e0...), on a rebuilt toolchain (K v7.1.337, Z3 4.13.3-1, clang/LLVM-15 15.0.6-4+b1, KEVM v1.0.921 source, blockchain-k-plugin @ 207ae512, crypto shim rebuilt with the keccak-multiblock fix). The frozen layers were not modified: `git diff` over `k/`, `python-gen/`, `phase1e/`, `phase1f/` against the Phase 1F tip is empty, and every Phase 1G extension is additive (`azCtx` / `linRecG` discipline; the KEVM authority adapter requires + imports the frozen modules).

1. Research question and answer

Question. Can SRW3 obtain and bind its authoritative state root and execution evidence from the execution/consensus layer WITHOUT circular trust, so that the commitment gate evaluates security obligations over evidence whose authority is grounded in the protocol/client execution process rather than merely supplied as an external assumption?

****Answer:** PARTIALLY PROVED — authority is formally INHERITABLE, and the inheritance is mechanized end-to-end; the chain's ROOT is protocol-given.**

The gate was extended (additively) so that acceptance (`valid-g`, 21 atomic conjuncts) requires, on top of the full Phase 1D-R1 + 1E + 1F certificate:

1. an AUTHORITY CERTIFICATE bound field-by-field to the record (payload digest, parent root, execution identity, child root, effect-trace digest) and to the security-policy version (`invalid-authbind`, `invalid-policy`);
2. the certificate's authority source to be an AUTHORIZED identity derived ONLY from the verification context — `chainId`, fork config, policy version, pinned proof descriptor — never from the certificate itself

- (invalid-authsrc, the definitional non-vacuity of IndependentAuthoritySource);
3. the authority relation to exist and to match the source domain, with the claimed level equal to the domain's level (invalid-authtype, invalid-authrel, invalid-authlevel);
 4. the authority chain to be NON-CIRCULAR: the certificate never cites itself (neither in its chain nor as its own source), every chain step carries an authorized identity, levels strictly decrease, and the chain terminates at a level-0 protocol root — consensus or policy (invalid-authcircle);
 5. the security context to be pinned (policyV, chainId, application set, interaction graph, lineage head) — the architectural bridge between execution validity and SRW3 security validity.

What this result IS NOT, stated plainly: the model's protocol roots (consensus identity for chainId; the pinned producer identity sets; the policy version's own authorization) are protocol-given INPUTS (A-G2/A-G3/A-G7). SRW3 verifies that authority chains terminate at a protocol root and never at themselves — authority is INHERITED by SRW3, not created. In a real deployment the consensus→execution boundary is the Engine API and the root becomes authoritative inside the consensus process: that last step is REQUIRES CLIENT/PROTOCOL SUPPORT (PROVED for the formal adapter at the model layer; the deployment gap is stated, not hidden). The Phase 1E/1F boundaries (A-E3 as the bare-anchor assumption; A-F3 trace authenticity at the abstract layer) are REPLACED at the Gate_G level by structured, checkable authority — the frozen verdicts themselves remain unchanged (frozen-layer discipline).

2. Starting state and environment

The session began from a sandbox reset (the 7th). The repo was re-cloned from GitHub; phase1e @ 8b26cb74 and phase1f @ b6c651e0 were verified on the fetched history before phase1g was derived (no history rewrite). The toolchain was rebuilt from the pinned recipe with two recorded deviations: (a) the K deb download truncated at 43/181 MB (resumed; dpkg-deb -c verified: 488 archive entries, 432 files); (b) libclang-common-15-dev is downloaded by rebuild_env_1d.sh but missing from its extract loop — stddef.h was absent for the shim build (extracted manually; the 1D recipe note anticipated the dependency). New setup script: scripts/phase1g/setup_env_1g.sh (clang path patch, syslibs shims, shim rebuild, env.sh). Symlinks are forbidden by the sandbox — the working repo was renamed to the canonical path instead (recorded).

The frozen Python layers re-verify transitively on every 1G case: the 1G Python suite imports and RUNS the full 1D→1E→1F chains (the b02 mutation re-demonstrates the 1F L14-ANCHOR catch).

3. The authority model (FROZEN)

phase1g/semantics/MODEL.md is the normative specification. Design points:

- **Four separations (never collapsed).** Evidence validity (Verify(root, evidence) = true), evidence authority (root = protocol/client-authoritative), execution validity (execution(payload, pre) = post), SRW3 security validity (obligations satisfied). The investigated relation: Authority → Authenticity → Execution validity → State/effect binding → Security validity → Commitment.
- **Authority domains and levels.** domConsensus/domPolicy at level 0 (protocol authorization roots), domExecution/domProof at level 1 (evidence producers); the certified object sits at level 2. Authority flows strictly from lower to higher.
- **AuthorityCertificate.** `azCert(payloadD, parentRoot, execlD, childRoot, effectD, rel, src, chain, policyV, proofPub)` with $rel \in \{1,2,3\}$:
1 = consensus-authorized payload executed by the authorized client (the Engine-API-modeled path), 2 = re-execution-derived (**Mode A**), 3 = proof-checked (**Mode B**). The source carries (domain, id, level); the chain carries the authority path above the source.
- **Authorized identities are CONTEXT-derived** (tags 0xA1..0xA4):
ConsensusId(chainId), ExecClientId(chainId, configD),
ProofSysId(authPT, authGuestD, authCfgD) (the descriptor PINNED in the context), PolicyAuthId(policyV). No check ever consumes the certificate to define what counts as authoritative — the §18 requirement at the definitional level.
- **NoCircularAuthority.** `certId ∉ chain-ids ∧ srcId ≠ certId ∧ AzChainOk(chain, srcLevel, ctx)`: strict level decrease, EVERY chain step authorized (context-derived checks — a producer cannot park an arbitrary claimed id and call it consensus), termination at an authorized level-0 protocol root (empty chain valid iff the source is itself level 0). Strict decrease + level-0 termination imply acyclicity. The fully self-referential certificate is a hash fixpoint — excluded mechanically by the own-id checks and computationally by A-G1.
- **SecurityContext.** `secCtx(policyV, chainId, appSetD, graphD, lineageHead)` — the policy binding is part of the accept condition

(invalid-policy); the policy's OWN authority is protocol/governance input (A-G7, disclosed; no in-model derivation — no infinite regress).

- **Record.** $\text{LinRecG} = (\text{LinRecF}, \text{azCertD}, \text{azPolicyV}); \text{`CanonCoreG} = \text{CanonCoreF} \parallel \text{azCertD} \parallel \text{policyV}_4 \text{`}$ — CanonCoreF a STRICT PREFIX; childE and childF carried unchanged (mechanized: AZR3/AZR4).
- **proofPub (Mode B public inputs).** `H}(\text{0x8D} \parallel \text{payloadD} \parallel \text{parentRoot} \parallel \text{chainId} \parallel \text{sched}_8 \parallel \text{spec}_4 \parallel \text{childRoot} \parallel \text{policyV}_4 \parallel \text{appSetD} \parallel \text{graphD} \parallel \text{lineageHead}) \text{` — beyond the EIP-8025-style bindings (payload identity, chain ID, schema/fork, child root), SRW3 adds the security-policy version, application set, interaction graph, and lineage head.
- **Domain tags** $\text{0x8C}/\text{0x8D}/\text{0x8E}/\text{0xA1}.. \text{0xA4}$ — collision-free against the frozen set (audited: the frozen tags are $\text{0x00}.. \text{0x03}$, $\text{0x11}.. \text{0x13}$, 0x1F , 0x57).

4. Gate_G (first-fail, total; 21 atomic conjuncts)

$\text{VerifyLineageG}(\text{recG}, \text{azCtx}, \text{tid}, \text{head})$: the full 1F chain fires first (all its verdicts unchanged), then the SEVEN new authority layers — invalid-authtype ($\text{rel} \notin \{1,2,3\}$), invalid-authrel ($\text{rel} \leftrightarrow \text{domain}$ mismatch), invalid-authlevel (claimed level $\neq$ domain level; CM-G3a), invalid-authcircle (NoCircularAuthority fails; CM-G1/2/3b — checked BEFORE source authorization so self-referencing certificates are diagnosed as circular even when their source could not be authorized), `invalid-authsrc` (source $\neq$ context-derived authorized identity; CM-G6/G10), invalid-authbind (certId recomputation + the five field bindings + the proofPub rule; CM-G4/5/7), invalid-policy (CM-G8) — and valid-g.

A-E3 replacement (the §7 experiment). Under valid-g:

$\text{anchor} == \text{stateRoot} == \text{cert.childRoot}$ where the certificate is non-circular and protocol-rooted — the anchor is no longer a bare assumption at the Gate_G level; it must CARRY authority. The frozen valid-a verdict is unchanged (the replacement applies to the commitment decision, which now requires valid-g).

5. Mechanized proofs (Haskell backend, kprove)

phase1g/proofs/ — 28 claims; 21 PROVED, 3 designed-fail, 4 disclosed (the per-claim raw outputs + exit codes are recorded verbatim):

Group	Claims	Outcome
Record binding: CanonCoreF strict prefix, childG binding, childE/childF invariance	AZR1-AZR4	4 PROVED

Group	Claims	Outcome
Gate_G branch-wise verdict exactness (each new verdict exactly its layer condition)	AZB1-AZB7	7 PROVED
Accept direction (the 21-conjunct accept $\Rightarrow$ valid-g, no acceptance slack)	AZBACC	1 PROVED
Authority-binding theorem classes under the full accept (payload/parent/execlD/effect/childRoot)	AZG1-AZG5	5 PROVED
Negative controls (designed fail: circular/unauthorized must never be accepted; vacuity probe)	NCG1-NCG3	3 designed NOT-PROVED (correct)
Single-implication forms over fully symbolic data	AZR5, AZNCS, AZCHS, AZCHG	4 NOT PROVED — DISCLOSED

The four disclosed attempts share the 1F EXGATE boundary: they require UNFOLDING a hash over a symbolic certificate (the premise-side AzCertBind0k contains AzCertId(C), which cannot evaluate for symbolic C — the NC1 keccak-injectivity precedent). Their CONTENT is carried by the proved claims: AZR5's five field equalities are AZG1-AZG5 (PROVED under the full accept) plus the digest binding consumed by AZBACC; the non-circularity content of AZNCS is consumed by AZBACC (the accept direction) and AZB4 (the verdict exactness); AZCHG's anchor half is carried by AZG5 + the frozen 1E anchor layer + the CM demonstrations (stale anchors rejected at L14-ANCH0R; child-root substitution rejected at invalid-authbind).

Non-vacuity: the accept premise is SATISFIABLE — valid-g is a permanent demo case for rel=1/2/3 on the LLVM backend with real keccak + secp256k1, mirrored by the Python suite; NCG3 (the vacuity probe) fails as designed.

6. The abstract demonstration suite (LLVM, real krypto)

srw3authz-demo.k — 8 partitioned programs (the concatenated suite exceeds the 3 GB sandbox as ONE term; the 1F precedent; no case skipped) —

7/7 PASS plus the byte program:

Case	Construction	Verdict
pos0	honest oracle record, rel=1 consensus-authorized	valid-g
pos1/pos2	lending threaded on R0's root (rel=1); liquidation with a Mode A certificate (rel=2, chain = [consensus root])	valid-g ×2
pos3	Mode B over R0 (rel=3, authorized proof descriptor, proofPub bound incl. policy/appSet/graph/lineageHead)	valid-g
g1 (CM-G1)	self-authorizing anchor: chain cites an unauthorized id as its consensus root	invalid-authcircle
g2 (CM-G2)	the source id is a CERTIFICATE-derived id (never a protocol identity)	invalid-authsrc
g3a (CM-G3a)	the client claims level 0	invalid-authlevel
g3b (CM-G3b)	client-sourced certificate with NO consensus chain	invalid-authcircle
g4 (CM-G4)	payload substitution	invalid-authbind

Case	Construction	Verdict
g5 (CM-G5)	parent-root substitution	invalid-authbind
g6 (CM-G6)	fork/config substitution: the client authorized under fork A, evaluated under fork B	invalid-authsrc
g7 (CM-G7)	child-root substitution	invalid-authbind
g8 (CM-G8)	policy substitution (cert policyV=2 vs SecCtx policyV=1)	invalid-policy
g9 (CM-G10)	proof from an unauthorized proof type	invalid-authsrc
g9b (CM-G10)	authorized proof source, proofPub does not recompute	invalid-authbind
g10	unknown authority relation (rel=4)	invalid-authtype
g11	rel↔domain mismatch	invalid-authrel
bal1	honest BAL: footprint + post-values bind	true
bal2 (CM-G9a)	omitted touched location — detected against the digest-bound trace	false (detected)
bal3 (CM-G9b)	the STALE-READ attack: BAL footprint=true ∧ writes=true while the ordered-trace evidence says invalid-effbind	THE FINDING
bal4 (CM-G9)	order blindness: XT2/XT2x share footprint and post-values; trace digests differ	BAL cannot see order

7. Python mirror and mutation harness

phase1g/python/ mirrors the K semantics exactly (authz_model.py) and re-runs the full case table (test_authz_verify.py: **25/25**) plus the hostile-mutation harness (authz_mutations.py: **18/18 as expected**) — the producer re-signs and re-digests everything forgeable (including arbitrary certificates with recomputed ids); rejections land at the documented first-fail layers (m01 digest flip → AUTHBIND; m02 self-chosen consensus id → AUTHSRC; m03 chain-cites-a-certificate → AUTHCIRCLE; m04 level-0 client → AUTHLEVEL; m05 no consensus chain → AUTHCIRCLE; m06b non-decreasing chain → AUTHCIRCLE; m07 foreign proofPub → AUTHBIND; m08 unauthorized proof type → AUTHSRC; m09 policy substitution → POLICY; m10 foreign execlId → AUTHBIND; m11 foreign childRoot → AUTHBIND; m12 rel/domain mismatch → AUTHREL; m13 unknown rel → AUTHTYPE; m14 foreign payload → AUTHBIND; m15 stale parentRoot → AUTHBIND). Documented boundaries: b01 (the honest rel=2 certificate is accepted — whether the source really is the protocol's client is the context-pinning assumption A-G3), b02 (the 1F fully-consistent hidden-write forgery is still caught by the FROZEN Level-2 anchor BEFORE any authority layer runs).

8. The authority adapter over real EVM execution (Tier G2)

k/kevm/srw3-authz-evm.k (+ demos; part5_kevm_authz.txt):

- **The certificate is constructed FROM THE RUN** — Mode A (rel=2): the source is the authorized execution client for (chainId, config), the authority chain carries the consensus root, and the certificate's effect digest binds the EXECUTION-DERIVED trace (the 1F instrumented tracer), so invalid-authbind forces presented-trace-digest == executed-trace-digest: at this layer the presented evidence IS the executed evidence.
- **Demos** (real keccak + secp256k1; commit/lineage atomicity preserved):
 - evm_authz_commit — honest run + adapter certificate: ****valid-g** → **COMMIT**** (the lineage record now carries the authority certificate);
 - evm_authz_selfauth — the SELF-AUTHORIZING variant (the client claims LEVEL 0, CM-G3a): **invalid-authlevel** → **REJECT + RESTORE** — the model does not accept "client output as unquestionable authority" even at the execution layer;
 - evm_authz_tamper — tampered presented trace: ****invalid-effdecl** → **REJECT + RESTORE**** (the FROZEN 1F chain diagnoses first — the layering discipline holds over real execution).
- **Memory-ceiling workarounds (documented, no case skipped).** (a) The adapter's certificate takes the F record as an ARGUMENT (projections only) — recomputing the Merkle build inside the cert OOM-killed the commit demo. (b) The KEVM harness evaluates the F chain ONCE and then the seven authority layers in isolation (#w3AzVerdict — the same verdict surface with the F half guaranteed by the dispatch; the integrated VerifyLineageG remains the canonical form, proved at the abstract layer and green in the abstract demos). The first-fail cascade over real EVM state is 9³ PE-chain evaluations — a sandbox-ceiling artifact, not a semantic one.
- Model chain identity #w3AzChainId = 0x00000001 (matches -cCHAINID=1; the KEVM config plumbing is not modified — fragment boundary).

9. Cross-layer byte consistency

scripts/phase1g/part6_crosslayer.py (part6_crosslayer_bytes.txt): ****10/10** byte-equality checks **MATCH**** across Python ↔ K-abstract (llvm, real krypto): cert body, certId, consensus/exec-client/proof-system/policy identities, proofPub, childG, CanonCoreG, BAL canon.

scripts/phase1g/part6b_kevm_cert.py (part6b_kevm_cert.txt): **7/7 MATCH** across Python ↔ the REAL KEVM run — the AUTHORITY CERTIFICATE of the actual execution is rebuilt in Python from first principles (payload, the #w3FParRoot parent root, the recomputed execution identity, the 1F anchored

child root, the EXECUTION-derived trace digest, the Mode A source, the consensus-rooted chain) and is byte-identical to the certificate the real run committed (the commit run's `linAzCertD == the dbg carrier's certId == the Python rebuild = a36cfab4...4b9fef`); the executed-trace digest == the presented-trace digest (4a3a1f19...); the authorized consensus/exec-client identities match. Three-layer byte consistency: ****Python ↔ K-abstract ↔ KEVM (real execution) — 17/17 MATCH****.

10. Prior-art positioning (targeted, live-fetched; `part7_prior_art.txt`)

Sources fetched this session (sha256 recorded in the transcript): EIP-8025 (Draft, Core), EIP-7928 (Last Call, Core), EIP-8159 (Last Call, Networking), the Engine API spec (`engine_newPayloadV1`).

Technique	What it already provides	The SRW3 delta
EIP-8025 execution proofs	opt-in consensus-layer stateless payload verification; public input binds <code>NewPayloadRequest</code> + chain ID + schema + validation result; verified proofs are "a supplementary validity signal, not a replacement for re-execution"	8025 does NOT make proof-checked authority load-bearing; no security-policy binding, no lineage, no authority structure. SRW3's <code>proofPub</code> additions (<code>policyV</code> , <code>appSetD</code> , <code>graphD</code> , <code>lineageHead</code>) and the <code>rel/level/chain</code> non-circularity are the delta; the proof math stays REQUIRES EXTERNAL PROOF SYSTEM (8025 is the natural host)
EIP-7928 BALs (enforced)	accounts + storage locations + post-execution values; tx-level write attribution; <code>storage_reads = read-only KEYS ONLY</code>	structurally confirms the 1G BAL findings: read OBSERVATIONS absent (the oracle-staleness class invisible), intra-tx same-slot order absent (F4 class invisible); BAL can serve as an independent footprint CROSS-CHECK (<code>bal2</code> : omission detected) but is NOT a security effect trace
EIP-8159 BAL exchange	p2p transport of BALs	networking only; no authority, no commitment mediation
Engine API	the consensus/execution boundary; <code>{status, latestValidHash}</code>	exactly the modeled level-0→1 edge; the root becomes authoritative through the consensus process; the smallest SRW3 insertion point that does NOT violate the separation is a post-execution hook exposing (<code>payload</code> , <code>parentRoot</code> , <code>result</code> , <code>childRoot</code> , <code>evidence</code>) — making a gate REJECT consensus-visible is REQUIRES CLIENT/PROTOCOL SUPPORT (Tier G3, not attempted)
Stateless witnesses / zkEVM	state-piece authentication; generic execution validity	the frozen standing distinction (generic validity ≠ SRW3 security validity) applies verbatim; complementary as a future Mode-B proof source

No surveyed system models **AUTHORITY** as distinct from authenticity with a non-circularity obligation. The comparison upgrades no evidence tier.

11. The two authority paths — comparison finding

Mode A (re-execution, rel=2) closes the 1F trace-authenticity boundary WHERE THE VERIFIER EXECUTES: demonstrated at Tier G2, where the certificate is derived from the run itself and the presented evidence must equal the executed evidence. **Mode B (proof-backed, rel=3)** delegates the execution

claim to an external proof system: SRW3 mechanizes the public-input binding and the source authorization; the proof math is **REQUIRES EXTERNAL PROOF SYSTEM**. **Neither mode eliminates A-G2**: both **INHERIT** authority from the consensus root of the chain (the Mode B certificate's chain also terminates at consensus). "Do not assume proof-backed execution is automatically more authoritative" — confirmed: Mode B's authority is exactly as rooted, and its execution claim is as strong as the proof system's soundness (A-G6), whereas Mode A's execution claim is as strong as the client's own re-execution inside the verifier's trust domain.

12. Consensus boundary (§9 questions, answered by the model + prior art)

Who authorizes the payload — the consensus layer (level 0; the Engine-API modeled edge). Who authorizes the execution context — the protocol pins the fork config and the authorized client identity per (chainId, config). Who determines the expected parent — the consensus chain (the payload's parentRoot binding). Who determines the expected resulting root — the execution client derives it; consensus makes it authoritative by authorizing the payload whose execution produced it. Who can invalidate — consensus (via fork choice), not the EL alone (Engine API: `{status: INVALID}` is a response, not a finality decision). At which layer does the root become authoritative — inside the consensus process (fork choice / finalization), which is exactly why the model's root is protocol-given (A-G2). At which layer may SRW3 legally reject — at its own commitment point: SRW3's **REJECT** is a refusal to commit security obligations, **NOT** a consensus invalidation (the separation is preserved by construction).

PHASE 1G RESULTS

Research question: can SRW3 obtain and bind its authoritative state root and execution evidence from the execution/consensus layer without circular trust?

Status: PARTIALLY PROVED – authority is formally INHERITABLE and the inheritance is mechanized end-to-end (21 kprove claims PROVED); the chain's ROOT is protocol-given (A-G2): the deployment-side grounding is REQUIRES CLIENT/PROTOCOL SUPPORT

Authority model: AuthzDomain(level 0: consensus/policy; level 1: execution/proof) + AuthorityCertificate(rel 1/2/3) + authority chain + SecurityContext; authorized identities CONTEXT-derived only

Authority graph: nodes = protocol roots, evidence producers, the certified object (level 2); edges = authorizes/derives/verifies/binds/commits; attestation carries NO authority; acyclicity = strict level decrease + level-0 termination (mechanized in AzChainOk/AzNonCircular)

Payload authority: PROVED BY K (AZG1: accept binds cert.payloadD == PayloadDigest(presented))

Parent-root authority: PROVED BY K (AZG2: cert.parentRoot == the threaded 1E parent root)

Execution authority: binding PROVED BY K (AZG3: cert.execId == the recomputed ExecutionId); result authority DEMONSTRATED BY KEVM (Mode A re-execution; presented == executed forced by authbind); Mode B REQUIRES EXTERNAL PROOF SYSTEM

Effect authority: PROVED BY K (AZG4: cert.effectD == the digest-bound ordered trace digest)

Child-root authority: PROVED BY K (AZG5: cert.childRoot == stateRoot; with the frozen anchor layer this is the A-E3 replacement at the Gate_G level)

Authority source: context-derived identities only (0xA1..0xA4); the certificate is never an input to its own authorization (definitional non-vacuity)

Authority verification: recomputation at every layer (certId, source identity, chain steps, level discipline, binding, policy)

Authority provenance: the chain terminates at an authorized level-0 protocol root; provenance beyond that root is protocol-given (A-G2) – REQUIRES CLIENT/PROTOCOL SUPPORT in deployment

Non-circularity theorem: AZB4 (verdict exactness) + AZBACC (accept consumes the non-circularity conjunct): PROVED BY K branch-wise; the single-form AZNCS disclosed (symbolic-hash boundary); the fully self-referential certificate is a hash fixpoint (mechanically excluded + computationally infeasible under A-G1)

G1: PROVED BY K (AZG1)
 G2: PROVED BY K (AZG2)
 G3: binding PROVED BY K (AZG3); result DEMONSTRATED BY KEVM (Mode A); Mode B REQUIRES EXTERNAL PROOF SYSTEM
 G4: PROVED BY K (AZG4)
 G5: PROVED BY K (AZG5)

G6: PROVED BY K (branch-wise: AZB4+AZBACC; the symbolic chain-induction form NOT YET MECHANIZED, disclosed)
 G7: PROVED BY K (AZR1-AZR4: strict prefix + childG binding + legacy invariance)
 G8: PROVED BY K (AZB1-AZB7 branch-wise + AZBACC accept direction; single form disclosed)
 G9: DEMONSTRATED (the 3-record authority chain, rel=1/1/2 + Mode B over R0, per-record certificates + lineage-head threading; the abstract composition claims are ground demos + the proved per-record certificate)
 CM-G1: REJECT (invalid-authcircle)
 CM-G2: REJECT (invalid-authsrc; the exact-fixpoint variant excluded by the own-id checks + A-G1)
 CM-G3: REJECT (a: invalid-authlevel; b: invalid-authcircle) – also demonstrated at the KEVM layer (evm_authz_selfauth)
 CM-G4: REJECT (invalid-authbind)
 CM-G5: REJECT (invalid-authbind)
 CM-G6: REJECT (invalid-authsrc – the client of fork A is not the authorized client under fork B)
 CM-G7: REJECT (invalid-authbind)
 CM-G8: REJECT (invalid-policy – the lineage pins the policy version)
 CM-G9: (a) omission DETECTED against the digest-bound trace (bal2); (b) the stale-read attack PASSES every BAL check while the ordered-trace evidence REJECTS it (bal3) – access list is NOT a complete security effect trace (also confirmed at the EIP-7928 spec level: storage_reads are keys only)
 CM-G10: REJECT (invalid-authsrc; proofPub mismatch is invalid-authbind – g9b)
 Phase 1E integration: DONE – the anchor must now CARRY a non-circular protocol-rooted certificate for the Gate_G decision (the A-E3 replacement); the frozen 1E verdicts unchanged
 Phase 1F integration: DONE – the full 1F chain fires first; the certificate binds the 1F execution identity and trace digest; at the KEVM layer the certificate's effect digest binds the EXECUTION-DERIVED trace (presented == executed)
 Lineage integration: DONE – LinRecG additive; CanonCoreF strict prefix; azCertD + policyV committed into the record (childG); legacy chains unchanged
 Gate integration: DONE – the commitment decision requires valid-g (21 conjuncts); no authoritative evidence = no security-valid commitment
 Access-list analysis: BAL = footprint + post-values (+tx-level write attribution); read observations and intra-tx same-slot order structurally absent; usable as an independent footprint CROSS-CHECK only; confirmed against EIP-7928's spec
 Execution-proof analysis: abstract ExecutionProofVerifier; public-input binding mechanized (incl. the SRW3 additions beyond EIP-8025); proof math REQUIRES EXTERNAL PROOF SYSTEM; 8025 currently treats proofs as supplementary – not load-bearing

KEVM result: 3/3 demos PASS – honest -> valid-g COMMIT;
 self-authorizing (level-0 client) ->
 invalid-authlevel REJECT+RESTORE; tampered ->
 invalid-effdecl REJECT+RESTORE (frozen 1F
 chain diagnoses first)
 Client-boundary result: the adapter IS the smallest authoritative
 integration point (post-execution state root
 + execution evidence -> authority statement);
 demonstrated, not enshrined
 Consensus-boundary result: modeled as the level-0 authorization root;
 making a gate REJECT consensus-visible is
 REQUIRES CLIENT/PROTOCOL SUPPORT
 Python/K consistency: 10/10 authority-layer byte checks MATCH
 (P <-> K-abstract, llvm real krypto) + 7/7
 Python <-> REAL KEVM RUN (the authority
 certificate of the actual execution rebuilt
 byte-identically in Python; the commit run's
 linAzCertID == the dbg certId == the rebuild)
 + 25/25 verdict cases + 18/18 mutations
 K/KEVM consistency: the same module chain (SRW3AUTHZ imported by
 the KEVM harness); the 1F tracer supplies the
 execution-derived trace; the authority fields
 evaluate identically (shared definitions)
 What is proved: 21 kprove claims – the record binding (prefix,
 childG, invariances), the branch-wise Gate_G
 certificate (7 verdicts exact + accept
 direction), the five authority-binding
 classes (G1/G2/G3-binding/G4/G5)
 What is demonstrated: the 3-record authority chain (rel=1/1/2 +
 Mode B), all CM-G attacks rejected at exact
 layers, the BAL findings, the KEVM adapter
 over real execution (valid-g COMMIT; self-
 authorization REJECTED at the EVM layer)
 What is assumed: A-G1 keccak collision resistance; A-G2 the
 consensus identity as the authorization root
 (the structured replacement of A-E3); A-G3
 the pinned producer identity sets; A-G7 the
 policy version's own authority (governance)
 What is refuted: "a certificate can carry its own authority" –
 every self-authorizing construction is
 rejected at an exact layer (CM-G1/2/3, NCG1/2
 designed-fail canaries); "access list =
 complete security effect trace" (CM-G9);
 "proof-backed authority is automatically
 stronger" (§11)
 What requires client/protocol support: the consensus-side grounding of A-G2
 (the Engine-API boundary); a consensus-visible
 commitment gate (Tier G3, not attempted);
 verifier re-execution as a service
 What requires an external proof system: Mode B's proof object (A-G6); the
 binding is mechanized, the math is not
 What remains open: the single-implication forms over symbolic
 data (AZR5/AZNC5/AZCH5/AZCHG – the symbolic-
 hash boundary, disclosed; content carried
 branch-wise); the tier-3 universal
 reconciliation (standing); generalizing the
 single-pinned identity sets to membership
 sets; symbolic chain-length induction

Authority-bound SRW3 conclusion: SRW3 can REQUIRE, CHECK, and COMMIT-ON
 inherited authority – every accepted
 commitment now carries a non-circular,
 context-authorized, field-bound authority
 certificate terminating at a protocol root –
 but SRW3 cannot CREATE the root's authority:
 the last step from "protocol-given" to
 "protocol-verified" is a client/protocol
 change, not a verifier-side one
 Artifact SHA-256: SRW3-Phase1G-Artifact-Bundle.zip + .sha256
 sidecar (MANIFEST 56 entries, round-trip
 ALL OK; the manifest carries every file's
 sha256 including this report)
 Git branch: phaselg (derived from phasel1 @ b6c651e0;
 no history rewrite of any milestone branch)
 Git commit: (this closure commit)
 Phase 1G completion verdict: COMPLETE – all 17 completion criteria
 evaluated below; no criterion silently waived

13. Completion criteria evaluation (§28, all 17)

#	Criterion	Verdict
1	Authority explicitly distinct from authenticity	SATISFIED (MODEL.md §2/§3; the four separations)
2	The authority chain formalized	SATISFIED (domains/levels/rel/chain/SecurityContext; §3)
3	At least one non-circular authority theorem mechanized or deliberately refuted	SATISFIED (branch-wise PROVED: AZB4+AZBACC; the single form disclosed; the refutation direction carried by NCG1/2 + CM-G1/2/3)
4	At least one authoritative-root path demonstrated	SATISFIED (pos0/pos1 rel=1: consensus-rooted certificates accepted; g6 fork-substitution rejected)
5	At least one authoritative-execution/effect path demonstrated	SATISFIED (pos2 rel=2 + the KEVM adapter: presented == executed forced; pos3 rel=3 binding)
6	Payload/parent/child/execution-context substitution attacks tested	SATISFIED (CM-G4/5/7 + CM-G6 fork + CM-G5 parent + F5/F6 inherited)
7	At least one self-authorizing/circular certificate attack rejected	SATISFIED (CM-G1/2/3a/3b + NCG1/2 canaries + the KEVM selfauth demo)
8	State authority connected to Phase 1E	SATISFIED (§4 A-E3 replacement; AZG5)
9	Execution/effect authority connected to Phase 1F	SATISFIED (§10; the certificate binds the 1F identity + trace digest)
10	Authority integrated into the lineage	SATISFIED (LinRecG; AZR1-AZR4)
11	Authority integrated into the commitment gate	SATISFIED (Gate_G; the KEVM commit now commits a G-record)
12	Python ↔ K consistency demonstrated	SATISFIED (10/10 + 7/7 bytes incl. the real KEVM run + 25/25 verdicts + 18/18 mutations)
13	KEVM binding demonstrated	SATISFIED (3/3 demos; shared module chain)
14	Current execution-proof and access-list prior art compared	SATISFIED (§10; live-fetched EIP-8025/7928/8159 + Engine API)
15	Real client/protocol changes explicitly identified	SATISFIED (§12: the Engine-API insertion point; consensus-visible rejection = REQUIRES CLIENT/PROTOCOL SUPPORT; Tier G3 not attempted)
16	No result silently assumes the authority relation it claims to prove	SATISFIED (A-G2/A-G3/A-G7 explicit; identities context-derived; the census in §14)
17	Full reproducible provenance package produced	SATISFIED (transcripts + pinned recipes + per-claim raw outputs + MANIFEST)

14. Assumption census (nothing hidden)

ID	Assumption	Classification
A-G1	Keccak256 collision resistance	REQUIRES CRYPTOGRAPHIC ASSUMPTION (inherits A-E1/A-F1)
A-G2	The consensus identity for chainId is the protocol's authorization root	PROTOCOL ASSUMPTION — the STRUCTURED replacement of A-E3; in deployment the Engine-API boundary: REQUIRES CLIENT/PROTOCOL SUPPORT

I D	Assumption	Classification
A - G 3	The authorized producer identity sets (the execution client per (chainId, config); the pinned proof descriptor) are protocol-pinned	CLIENT/PROTOCOL ASSUMPTION
A - G 4	Canonical-encoding injectivity (cert/step/sec canon)	STRUCTURAL (inherits A-F2)
A - G 5	Fragment field bounds (1F §2 unchanged)	explicit REFUSE
A - G 6	Proof-system soundness for Mode B	REQUIRES EXTERNAL PROOF SYSTEM
A - G 7	The security policy version's own authority	PROTOCOL/GOVERNANCE ASSUMPTION (disclosed; no in-model derivation)

Inherited unchanged: A-E2/A-E4/A-E5, A-F1..A-F5, the 1D census. A-E3 is SUPERSEDED at the Gate_G level (§4). Implementation assumptions recorded in the PROVENANCE (the KEVM memory-ceiling factorization; the model chainId constant; the partitioned demo programs).

15. Artifact index

Artifact	Path
Model spec (FROZEN)	phase1g/semantics/MODEL.md
K semantics (additive)	phase1g/semantics/srw3authz.k + demo srw3authz-demo.k
K proofs + negative controls	phase1g/proofs/*.k
Python reference + suites	phase1g/python/{authz_model,test_authz_verify,authz_mutations}.py
KEVM authority adapter + demos	k/kevm/srw3-authz-evm.k, srw3-authz-evm-demos.k, k/kevm/demos/evm_authz_*.srw3evm
Transcripts (Parts 1-7)	phase1g/transcripts/
Cross-layer + prior-art scripts	scripts/phase1g/
Report	phase1g/report/SRW3-Phase1G-Authority-Report.{md,pdf}
Provenance	phase1g/provenance/PROVENANCE.md